DATA ANALYSIS REPORT

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Project: Healthcare Patient Engagement & Revenue Analysis

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Introduction

This report presents an end-to-end analysis of healthcare patient data, focusing on patient engagement, revenue generation, and operational performance across departments, doctors, and U.S. states. The dataset simulates hospital records including patient visits, appointment outcomes, and financial performance.

Purpose: To analyze patient behavior, identify revenue drivers, and evaluate performance across different states to support data-driven decision-making.

Objectives:

- Analyze patient engagement and retention using SQL
- Identify revenue trends and performance patterns
- Evaluate performance across U.S. states, departments, and doctors
- Visualize insights using Power BI dashboards

Dataset Description

Source: Simulated dataset generated in Excel and stored in MySQL

Fields:

- Patient_ID
- Visit_ID
- Full Name
- Gender
- Age
- Age Group
- Department
- Doctor
- Visit_Date
- Month-Year
- State (U.S. State)
- Appointment_Status (Completed / No-show)
- Revenue
- Patient Type (New / Returning)

Methodology

- Excel – Data generation and cleaning
- MySQL – Data storage and querying
- Power BI – Visualization and dashboard creation

Data Cleaning:

- Removed duplicate records
- Standardized categorical values (Gender, Appointment_Status)
- Handled missing values
- Created derived fields (Month-Year, Age Group, Patient Type)

Use of AI in Analysis

Artificial Intelligence (AI) tools were utilized to assist in generating and validating SQL queries and DAX calculations.

AI support was used to:

- Suggest optimized SQL queries for aggregations (e.g., revenue by state, patient retention)
- Assist in writing advanced SQL functions such as window functions (RANK, LAG) for ranking and trend analysis
- Help debug query errors and refine logic
- Provide guidance in structuring Power BI DAX measures

All AI-assisted outputs were reviewed and validated to ensure accuracy and alignment with analytical objectives.

SQL Analysis

Revenue by State

```
SELECT State, SUM(Revenue) AS Total_Revenue
FROM patient_data
WHERE Appointment_Status = 'Completed'
GROUP BY State
ORDER BY Total_Revenue DESC;
```

Identifies top-performing states based on revenue.

Monthly Revenue Trend

```
SELECT
    DATE_FORMAT(Visit_Date, '%Y-%m') AS Month,
    SUM(Revenue) AS Monthly_Revenue
FROM patient_data
WHERE Appointment_Status = 'Completed'
GROUP BY DATE_FORMAT(Visit_Date, '%Y-%m')
ORDER BY DATE_FORMAT(Visit_Date, '%Y-%m');
```

Tracks revenue growth over time.

Returning Patients

```
SELECT COUNT(*) AS Returning_Patients
FROM (
    SELECT Patient_ID
    FROM patient_data
    GROUP BY Patient_ID
    HAVING COUNT(Visit_ID) > 1
) AS sub;
```

Measures patient retention.

Top Doctors by Revenue

```
SELECT Doctor, SUM(Revenue) AS Total_Revenue
FROM patient_data
WHERE Appointment_Status = 'Completed'
GROUP BY Doctor
ORDER BY Total_Revenue DESC;
```

Highlights top-performing doctors.

No-show Analysis by State

```
SELECT
    State,
    COUNT(Appointment_status) as no_show
FROM patient_data
WHERE appointent_status = ‘No-Show’
GROUP BY state;
```

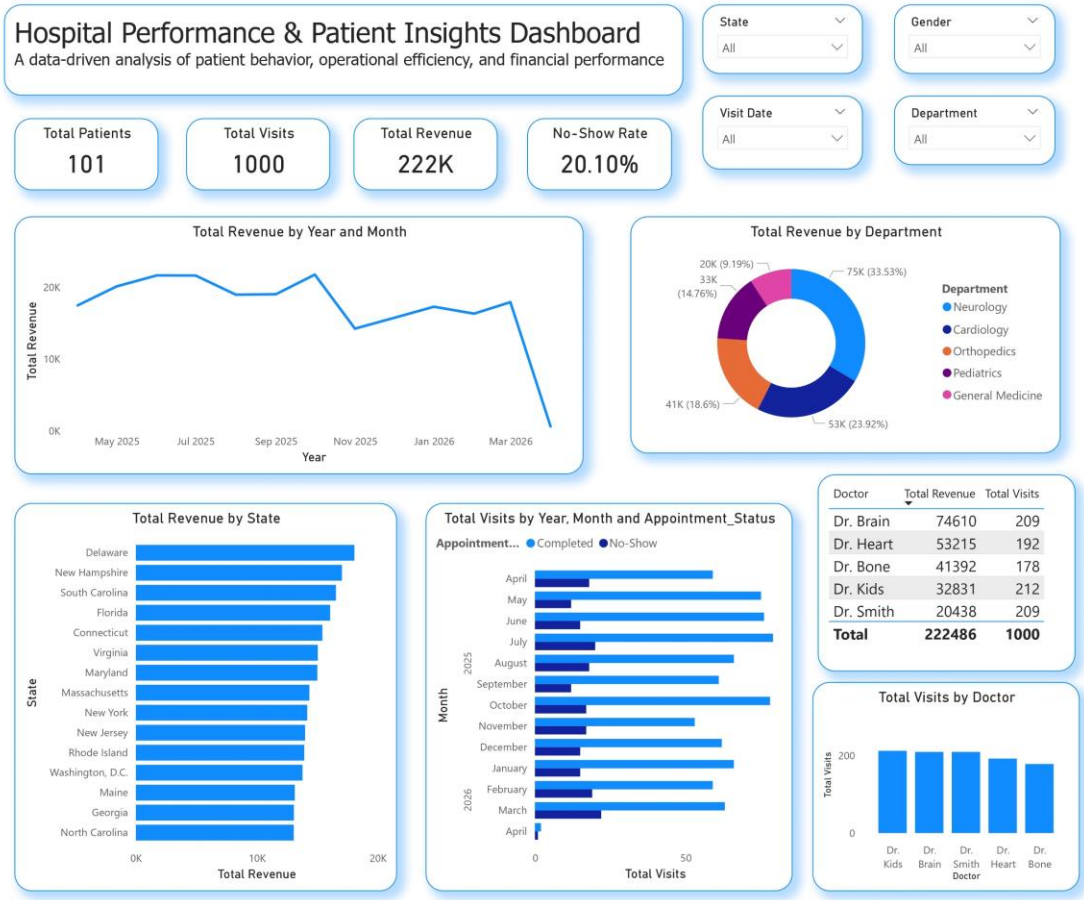
Evaluates appointment reliability across states.

Data Visualization

An interactive Power BI dashboard was developed to present insights clearly and effectively.

Key Visuals Included:

- KPI Cards: Total Patients, Total Revenue, Total Visits, No-show Rate
- Line Chart: Monthly Revenue Trend
- Donut Chart: Revenue by Department
- Bar Chart: Revenue by State
- Stacked Column Chart: Appointment Status Trends
- Doctor Performance Chart: Top 5 Doctors by Revenue



Findings / Insights

- Revenue shows a steady upward trend over time
- Certain departments contribute the majority of total revenue
- Patient retention is observed through repeat visits
- No-show rates vary across states, indicating operational inefficiencies
- Top-performing doctors generate a significant portion of revenue
- Some states outperform others in both patient volume and revenue

Conclusion

The analysis highlights key drivers of healthcare performance, including department efficiency, patient retention, and doctor productivity across different U.S. states.

Reducing no-show rates and focusing on high-performing states and departments can improve operational efficiency and revenue outcomes.

The integration of SQL and Power BI provides a strong foundation for scalable and data-driven healthcare analytics.

Appendix

- Full SQL query scripts
- Power BI dashboard file (.pbix)
- Excel dataset
- Supporting visuals and screenshots